The Transformer

Problem Setup

In previous chapters we built models that predict a continuous output $y \in \mathbb{R}$ from an input $\mathbf{x}$. Now consider the case where the input is a **sequence of discrete tokens** $w_1, w_2, \ldots, w_T$, each drawn from a finite vocabulary $\mathcal{V} = \{1, 2, \ldots, V\}$, and the goal is to predict the next token $w_{T+1} \in \mathcal{V}$.

By the chain rule of probability, the joint distribution of any sequence factorizes as:

$$p(w_1, w_2, \ldots, w_T) = \prod_{t=1}^{T} p(w_t \mid w_1, \ldots, w_{t-1})$$

So modelling sequences reduces entirely to modelling the conditional $p(w_t \mid w_{1:t-1})$. This is called **language modelling**. We seek a parametric function $f_\Theta : \mathcal{V}^* \to [0,1]^V$ such that:

$$p(w_t \mid w_1, \ldots, w_{t-1}) \approx f_\Theta(w_1, \ldots, w_{t-1})_{w_t}$$

where $f_\Theta(\cdot)_{w_t}$ denotes the w_t-th component of the output vector.

Problem. The vocabulary $\mathcal{V}$ of a typical model contains $V \approx 50{,}000$ tokens. A sequence of length T lives in a space of size V^T; we cannot enumerate all possible contexts. The classical N-gram model approximates $p(w_t \mid w_{1:t-1})$ by $p(w_t \mid w_{t-N+1:t-1})$, but this destroys dependencies beyond the window of N tokens.

The **transformer** is the architecture we use for f_Θ. It maps a variable-length sequence of token indices to a probability distribution over $\mathcal{V}$, and it can attend to the *entire* preceding context. We derive it below, following the original architecture (Vaswani et al., 2017) component by component.

Roadmap

An input passes through the following stages:

Stage	Component	Section
1	Token embedding	§2
2	Positional encoding (added to embedding)	§3
3	Masked multi-head attention	§4–6
4	Add & LayerNorm	§7
5	Feed-forward network	§8
6	Add & LayerNorm	§7
7	Repeat stages 3–6 $\times L$	§9
8	Final LayerNorm $\to$ Linear $\to$ Softmax	§10

Input Embedding

The input to the model is an integer $w_t \in \{1, \ldots, V\}$. Integers are discrete; we cannot differentiate with respect to them. We need a continuous representation.

Definition 1 (Embedding Matrix)**.** *Let $\mathbf{E} \in \mathbb{R}^{V \times d}$ be the **embedding matrix**, where d is the **model dimension**. The embedding of token w_t is the w_t-th row of $\mathbf{E}$:*

$$\mathbf{e}_t := \mathbf{E}[w_t] \in \mathbb{R}^{1 \times d}$$

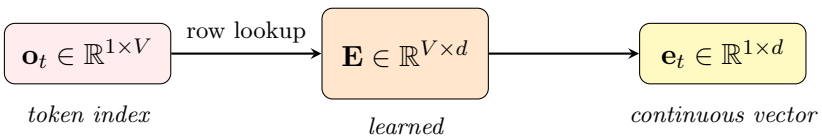
An equivalent viewpoint: represent the token as a one-hot vector $\mathbf{o}_t \in \mathbb{R}^{1 \times V}$ with a single 1 at position w_t and zeros elsewhere. Then:

$$\mathbf{e}_t = \mathbf{o}_t \mathbf{E}$$

Since $\mathbf{o}_t$ has exactly one nonzero entry, this matrix multiply reduces to a row lookup. The matrix $\mathbf{E}$ is a **learned parameter**: it is updated during training via backpropagation. Tokens that play similar roles in context get pushed to have similar embeddings.

For a full sequence of N tokens, we stack the embeddings into a matrix:

$$\mathbf{X}_{\text{emb}} = \begin{pmatrix} \mathbf{E}[w_1] \\ \vdots \\ \mathbf{E}[w_N] \end{pmatrix} \in \mathbb{R}^{N \times d}$$



Positional Encoding

The embedding $\mathbf{E}[w_t]$ depends only on the token identity, not on its position t in the sequence. But position matters: "dog bites man" and "man bites dog" contain the same tokens. We need to inject positional information.

The simplest approach: define a matrix $\mathbf{P} \in \mathbb{R}^{N_{\max} \times d}$ where row t is the positional encoding of position t. We **add** it to the token embedding:

$$\mathbf{x}_t = \mathbf{E}[w_t] + \mathbf{P}[t] \in \mathbb{R}^{1 \times d}$$

The matrix $\mathbf{P}$ can be learned (like $\mathbf{E}$), or it can be fixed. Vaswani et al. (2017) used a deterministic sinusoidal encoding:

$$\mathbf{P}[t,\, 2k] = \sin\left(\frac{t}{10000^{2k/d}}\right) \tag{1}$$

$$\mathbf{P}[t,\, 2k+1] = \cos\left(\frac{t}{10000^{2k/d}}\right) \tag{2}$$

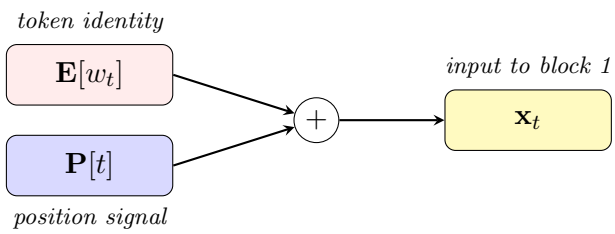
for $k = 0, 1, \ldots, d/2 - 1$. Each pair of dimensions oscillates at a different frequency, giving each position a unique signature.

Remark 2. *The positional encoding is **added**, not concatenated. Both $\mathbf{E}[w_t]$ and $\mathbf{P}[t]$ have shape $[1 \times d]$, so the sum $\mathbf{x}_t$ also has shape $[1 \times d]$. This preserves the model dimension throughout the architecture.*

For the full sequence:

$$\mathbf{X} = \mathbf{X}_{\text{emb}} + \mathbf{P}_{1:N} \in \mathbb{R}^{N \times d}$$

This is the input to the first transformer block.



Attention: Motivation

The next block is **Masked Multi-Head Attention**. This is the core of the transformer. Before deriving the full mechanism, we motivate it.

Consider the sentence: "The chicken didn't cross the road because **it** was too tired." The word "it" refers to "chicken". In "...because **it** was too wide", it refers to "road". To compute a useful representation of "it", we need to look at the other tokens and determine which ones are relevant.

Goal. Given a token at position i with representation $\mathbf{x}_i \in \mathbb{R}^{1 \times d}$, compute a new representation $\mathbf{a}_i \in \mathbb{R}^{1 \times d}$ that incorporates information from all preceding positions $j \leq i$.

The simplest version: take a weighted sum of the representations of all preceding tokens, where the weights reflect how relevant each token is to position i:

$$\mathbf{a}_i = \sum_{j \leq i} \alpha_{ij} \, \mathbf{x}_j, \qquad \alpha_{ij} \geq 0, \qquad \sum_{j \leq i} \alpha_{ij} = 1$$

The constraint $j \leq i$ is the **causal mask**: token i cannot look at future tokens, since at generation time those tokens have not been produced yet.

How should we compute α_{ij}? A natural choice is to use the dot product as a similarity measure, then normalize with a softmax:

$$\alpha_{ij} = \frac{\exp(\mathbf{x}_i \cdot \mathbf{x}_j)}{\sum_{k \leq i} \exp(\mathbf{x}_i \cdot \mathbf{x}_k)}$$

Problem. In this formulation, each $\mathbf{x}_i$ plays three roles simultaneously: the "query" (what am I looking for?), the "key" (what do I contain?), and the "value" (what information do I provide?). Conflating three distinct functions into one vector limits expressiveness.

Queries, Keys, and Values

To separate the three roles, we introduce three learned linear projections.

Definition 3 (Query, Key, Value)**.** *Given input* $\mathbf{x}_i \in \mathbb{R}^{1\times d}$*, define:*

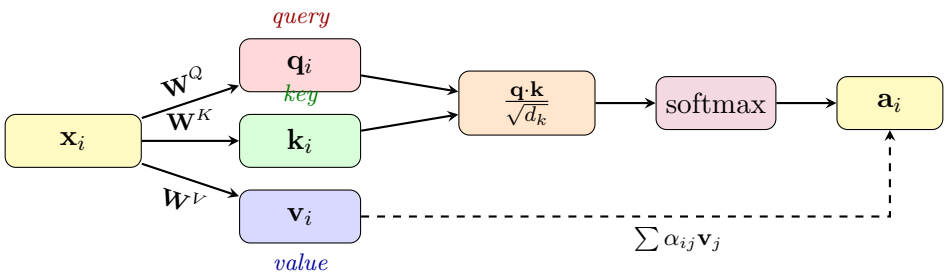
$$\mathbf{q}_i = \mathbf{x}_i\,\mathbf{W}^Q \in \mathbb{R}^{1\times d_k} \tag{3}$$

$$\mathbf{k}_i = \mathbf{x}_i\,\mathbf{W}^K \in \mathbb{R}^{1\times d_k} \tag{4}$$

$$\mathbf{v}_i = \mathbf{x}_i\,\mathbf{W}^V \in \mathbb{R}^{1\times d_v} \tag{5}$$

where $\mathbf{W}^Q \in \mathbb{R}^{d\times d_k}$*,* $\mathbf{W}^K \in \mathbb{R}^{d\times d_k}$*,* $\mathbf{W}^V \in \mathbb{R}^{d\times d_v}$ *are learned weight matrices.*

The **query** $\mathbf{q}_i$ asks "what am I looking for?". The **key** $\mathbf{k}_j$ answers "what do I contain?". The **value** $\mathbf{v}_j$ is "what information do I provide if selected?". Attention scores are computed between queries and keys; values are what gets summed.



Scaled Dot-Product Attention

The score between positions i and j is:

$$\text{score}(\mathbf{x}_i, \mathbf{x}_j) = \frac{\mathbf{q}_i \cdot \mathbf{k}_j}{\sqrt{d_k}}$$

Proposition 4 (Scaling preserves unit variance)**.** *If the entries of* $\mathbf{q}$ *and* $\mathbf{k}$ *are i.i.d. with mean* 0 *and variance* 1*, then:*

$$\mathbb{E}[\mathbf{q}_i \cdot \mathbf{k}_j] = 0, \qquad Var(\mathbf{q}_i \cdot \mathbf{k}_j) = d_k$$

Proof. The dot product $\mathbf{q}_i \cdot \mathbf{k}_j = \sum_{m=1}^{d_k} q_{im}k_{jm}$ is a sum of d_k independent terms. By independence, $\mathbb{E}[q_{im}k_{jm}] = \mathbb{E}[q_{im}]\mathbb{E}[k_{jm}] = 0$. For the variance:

$$\text{Var}(q_{im}k_{jm}) = \mathbb{E}[q_{im}^2 k_{jm}^2] - (\mathbb{E}[q_{im}k_{jm}])^2 = \mathbb{E}[q_{im}^2]\mathbb{E}[k_{jm}^2] - 0 = 1$$

Summing d_k such terms gives $\text{Var}(\mathbf{q}_i \cdot \mathbf{k}_j) = d_k$. □

So the standard deviation of the raw dot product grows as $\sqrt{d_k}$. For large d_k, the dot products become large in magnitude, pushing the softmax into saturation (outputs near 0 or 1) where gradients vanish. Dividing by $\sqrt{d_k}$ normalizes the variance back to 1.

Putting It Together

The attention weight from position i to position j is:

$$\alpha_{ij} = \frac{\exp\left(\mathbf{q}_i \cdot \mathbf{k}_j/\sqrt{d_k}\right)}{\sum_{m\le i}\exp\left(\mathbf{q}_i \cdot \mathbf{k}_m/\sqrt{d_k}\right)} \qquad \forall\, j \le i$$

The attention output is the weighted sum of values:

$$\text{head}_i = \sum_{j \le i} \alpha_{ij}\, \mathbf{v}_j \in \mathbb{R}^{1 \times d_v}$$

Finally, we apply an output projection $\mathbf{W}^O \in \mathbb{R}^{d_v \times d}$ to map back to the model dimension:

$$\mathbf{a}_i = \text{head}_i\, \mathbf{W}^O \in \mathbb{R}^{1 \times d}$$

Shape Summary

Object	**Shape**
Input $\mathbf{x}_i$	$[1 \times d]$
$\mathbf{W}^Q, \mathbf{W}^K$	$[d \times d_k]$
$\mathbf{W}^V$	$[d \times d_v]$
$\mathbf{q}_i, \mathbf{k}_j$	$[1 \times d_k]$
$\mathbf{v}_j$	$[1 \times d_v]$
$\mathbf{q}_i \cdot \mathbf{k}_j$	scalar
α_{ij}	scalar
head_i	$[1 \times d_v]$
$\mathbf{W}^O$	$[d_v \times d]$
$\mathbf{a}_i$	$[1 \times d]$

The input and output both have dimension d. This is by design: it means attention blocks can be stacked.

Parallelization via Matrices

The computation above describes a single output $\mathbf{a}_i$. For all N positions simultaneously, we pack the embeddings into $\mathbf{X} \in \mathbb{R}^{N \times d}$ and compute:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}^Q, \quad \mathbf{K} = \mathbf{X}\mathbf{W}^K, \quad \mathbf{V} = \mathbf{X}\mathbf{W}^V$$

All N^2 pairwise scores are computed in one matrix multiply:

$$\mathbf{S} = \frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} \in \mathbb{R}^{N \times N}$$

The entry S_{ij} is the score from position i to position j.

The Causal Mask

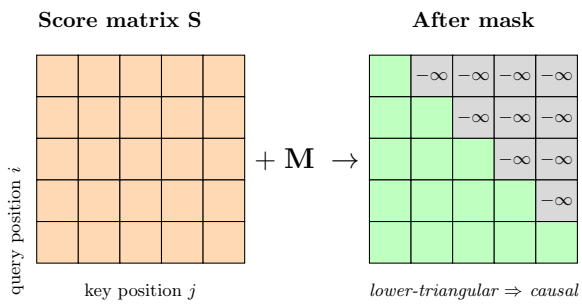
The matrix $\mathbf{S}$ contains scores for all pairs (i, j), including $j > i$ (future tokens). To enforce causality, we add a mask matrix $\mathbf{M} \in \mathbb{R}^{N \times N}$:

$$M_{ij} = \begin{cases} 0 & \text{if } j \le i \\ -\infty & \text{if } j > i \end{cases}$$

Since $e^{-\infty} = 0$, the softmax zeroes out all future positions. The full masked attention is:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} + \mathbf{M}\right)\mathbf{V} \in \mathbb{R}^{N \times d_v}$$

The resulting weight matrix is lower-triangular: position i attends only to positions $1, \ldots, i$.



Computational cost. The product $\mathbf{QK}^\top$ costs $O(N^2 d_k)$. Attention is therefore **quadratic** in the sequence length N.

Multi-Head Attention

A single set of projections $(\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V)$ can learn one type of attention pattern. But tokens relate to each other in many ways: syntactic agreement, coreference, semantic similarity. To capture multiple types of relationships simultaneously, we use **multiple heads**.

Definition 5 (Multi-Head Attention)**.** *Given A attention heads, each head $c \in \{1, \ldots, A\}$ has its own weight matrices $\mathbf{W}^{Qc}, \mathbf{W}^{Kc}, \mathbf{W}^{Vc}$. Each head computes:*

$$\mathit{head}^c = \mathit{Attention}\big(\mathbf{XW}^{Qc},\ \mathbf{XW}^{Kc},\ \mathbf{XW}^{Vc}\big) \in \mathbb{R}^{N \times d_v}$$

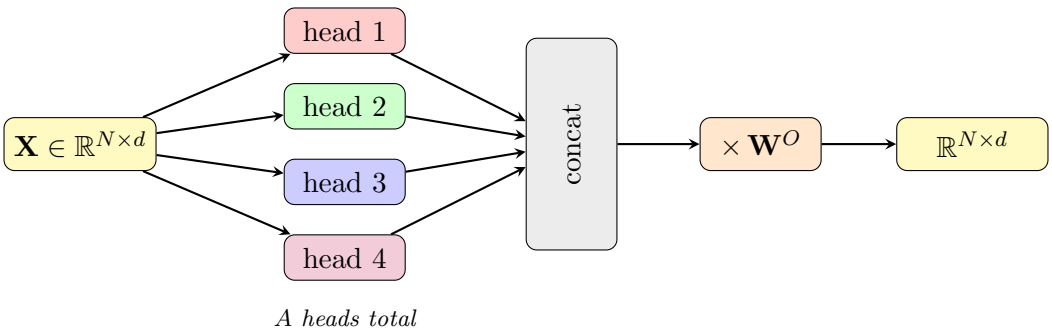
The heads are concatenated and projected:

$$\mathit{MultiHead}(\mathbf{X}) = \big[\mathit{head}^1 \parallel \mathit{head}^2 \parallel \cdots \parallel \mathit{head}^A\big]\, \mathbf{W}^O$$

where $\parallel$ denotes concatenation along the last axis and $\mathbf{W}^O \in \mathbb{R}^{Ad_v \times d}$.

Shapes. Each head output is $[N \times d_v]$. Concatenating A heads gives $[N \times Ad_v]$. Multiplying by $\mathbf{W}^O$ gives $[N \times d]$. It is standard to set $d_k = d_v = d/A$ so that $Ad_v = d$ and the total parameter count equals that of a single head with full dimension d.

In the original transformer: $d = 512$, $A = 8$, $d_k = d_v = 64$.



Remark 6. *The multi-head mechanism divides the d-dimensional representation into A subspaces of dimension d/A each. Each head learns an independent attention pattern in its subspace. This is strictly more expressive than a single head, because a single head must average over all types of relationships.*

Add & Norm

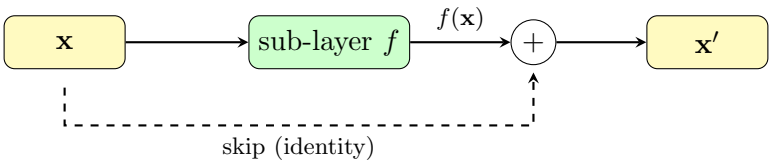
The next component (appearing after each sub-layer) is "Add & Norm". This combines two mechanisms: a **residual connection** and **layer normalization**.

Residual Connections

Given a sub-layer function f (either attention or feedforward), the residual connection is:

$$\text{output} = f(\mathbf{x}) + \mathbf{x}$$

This creates a direct path for the input to flow through unchanged, while the sub-layer only needs to learn a *correction*. This addresses the vanishing gradient problem: the gradient of $\mathbf{x} + f(\mathbf{x})$ with respect to $\mathbf{x}$ is $\mathbf{I} + \frac{\partial f}{\partial \mathbf{x}}$, which is always at least the identity.



Layer Normalization

Given a vector $\mathbf{x} \in \mathbb{R}^d$, layer normalization (Ba et al., 2016) normalizes it to zero mean and unit variance, then applies a learned affine transformation.

Step 1. Compute the mean and standard deviation over the d components of $\mathbf{x}$:

$$\mu = \frac{1}{d}\sum_{i=1}^{d} x_i \tag{6}$$

$$\sigma = \sqrt{\frac{1}{d}\sum_{i=1}^{d}(x_i - \mu)^2} \tag{7}$$

Step 2. Normalize and apply learned parameters $\gamma, \beta \in \mathbb{R}^d$:

$$\text{LayerNorm}(\mathbf{x}) = \gamma \odot \frac{\mathbf{x} - \mu}{\sigma + \epsilon} + \beta$$

where $\odot$ is element-wise multiplication and $\epsilon > 0$ is a small constant for numerical stability. The parameters γ (gain) and β (offset) are learned during training.

Remark 7. *Layer normalization is applied to a* ***single token's vector****, not across the batch or across positions. It is the z-score from statistics, applied dimension-wise to one embedding.*

Order of Operations

In the original transformer (Vaswani et al., 2017), the order is sub-layer → add → norm ("post-norm"). Most modern implementations use the "pre-norm" variant: norm → sub-layer → add. We write the pre-norm version, since it is now standard:

$$\mathbf{z} = \text{LayerNorm}(\mathbf{x}) \tag{8}$$

$$\mathbf{x}' = f(\mathbf{z}) + \mathbf{x} \tag{9}$$

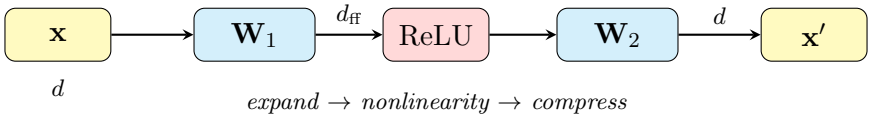
Feed-Forward Network

After the first Add & Norm, the diagram shows a **Feed Forward** block. This is a standard two-layer fully connected network, applied independently and identically to each token position:

$$\mathrm{FFN}(\mathbf{x}) = \mathrm{ReLU}(\mathbf{x}\,\mathbf{W}_1 + \mathbf{b}_1)\,\mathbf{W}_2 + \mathbf{b}_2$$

where $\mathbf{W}_1 \in \mathbb{R}^{d \times d_{\mathrm{ff}}}$, $\mathbf{W}_2 \in \mathbb{R}^{d_{\mathrm{ff}} \times d}$, and $\mathrm{ReLU}(z) = \max(0, z)$.

The hidden dimension d_{ff} is typically $4d$. In the original transformer, $d = 512$ and $d_{\mathrm{ff}} = 2048$.



Key point. The FFN processes each position independently: it sees only the d-dimensional vector at one position, with no interaction between positions. The attention layer is the only component that mixes information across positions.

After the FFN, there is another Add & Norm (the second one in each block).

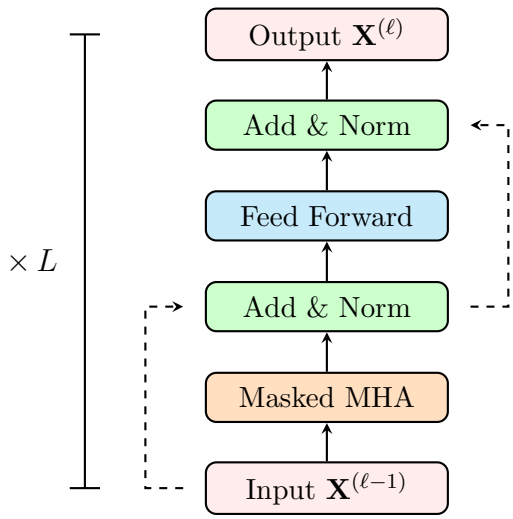
Stacking: L Blocks

The block (Masked Multi-Head Attention → Add & Norm → Feed Forward → Add & Norm) is repeated L times. In pre-norm convention, one block computes:

$$\mathbf{O} = \mathbf{X} + \mathrm{MultiHead}(\mathrm{LayerNorm}(\mathbf{X})) \tag{10}$$

$$\mathbf{H} = \mathbf{O} + \mathrm{FFN}(\mathrm{LayerNorm}(\mathbf{O})) \tag{11}$$

The output $\mathbf{H} \in \mathbb{R}^{N \times d}$ has the same shape as the input $\mathbf{X} \in \mathbb{R}^{N \times d}$. This is what makes stacking possible: the output of block ℓ is the input to block $\ell + 1$.



Typical depth: $L = 12$ for small models (GPT-2), $L = 96$ for large models (GPT-3), $L = 126$ for Llama 3.1 405B.

At the very top of the final block there is one additional LayerNorm applied to $\mathbf{X}^{(L)}$ before passing to the output stage.

Linear + Softmax (The Output Head)

The output of the last transformer block, after the final LayerNorm, is a matrix $\mathbf{H} \in \mathbb{R}^{N \times d}$. We take the row corresponding to the last token, $\mathbf{h}_N \in \mathbb{R}^{1 \times d}$, and map it to a probability distribution over the vocabulary $\mathcal{V}$.

The Linear Layer (Unembedding)

We need to go from $\mathbb{R}^d \to \mathbb{R}^V$. A natural choice: reuse the embedding matrix $\mathbf{E} \in \mathbb{R}^{V \times d}$ from §2, transposed. This is called **weight tying**.

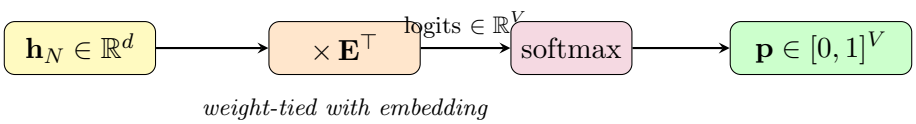
The **logit** (unnormalized score) for each token $k \in \mathcal{V}$ is:

$$u_k = \mathbf{h}_N \cdot \mathbf{E}[k]$$

In matrix form:

$$\mathbf{u} = \mathbf{h}_N \, \mathbf{E}^\top \in \mathbb{R}^{1 \times V}$$

The score for token k is the dot product between the contextual representation $\mathbf{h}_N$ and the embedding of token k. Tokens whose embeddings are "close" to $\mathbf{h}_N$ receive high scores.



Remark 8. *Weight tying means* $\mathbf{E}$ *must simultaneously be a good input representation (mapping tokens to* $\mathbb{R}^d$*) and a good output projection (mapping* $\mathbb{R}^d$ *back to token scores). This halves the number of vocabulary-related parameters and serves as a regularizer.*

Softmax

The logits $\mathbf{u} \in \mathbb{R}^{1 \times V}$ are real-valued and unnormalized. We convert them to a probability distribution using the softmax function:

$$p(w_{N+1} = k \mid w_{1:N}) = \frac{\exp(u_k)}{\sum_{j=1}^{V} \exp(u_j)}$$

This satisfies $p_k > 0$ for all k and $\sum_k p_k = 1$.

Numerical stability. In practice we compute $\mathrm{softmax}(\mathbf{u} - \max_k u_k)$. Subtracting the maximum does not change the output (the constant cancels in numerator and denominator) but prevents overflow in the exponential.

Training

The model is trained by **maximum likelihood**, which is equivalent to minimizing the **cross-entropy loss**.

At position t, the model outputs a distribution $\hat{p}(\cdot \mid w_{1:t})$ over $\mathcal{V}$. The true next token is w_{t+1}, which defines a one-hot distribution. The cross-entropy between them is:

$$\mathcal{L}_t = -\log \hat{p}(w_{t+1} \mid w_{1:t})$$

For a sequence of T tokens, we average:

$$\mathcal{L} = -\frac{1}{T} \sum_{t=1}^{T} \log \hat{p}(w_{t+1} \mid w_{1:t})$$

This is minimized via gradient descent (typically Adam) with backpropagation through all L transformer blocks.

Key advantage over RNNs. Thanks to the causal mask, the loss at every position t can be computed **in parallel** during training. The entire sequence is processed in a single forward pass. This is why transformers are faster to train than recurrent models, despite having quadratic attention cost.

Generation (Inference)

Given a trained model and a prompt $w_{1:N}$, we generate text by repeated next-token prediction:

1. Set $t \leftarrow N$.
2. **Repeat:**
   (a) Compute $\mathbf{h}_t^{(L)} = f_\Theta(w_{1:t})$ (forward pass)
   (b) $\mathbf{u} \leftarrow \mathbf{h}_t^{(L)} \mathbf{E}^\top$ (logits)
   (c) $\mathbf{p} \leftarrow \mathrm{softmax}(\mathbf{u})$ (probabilities)
   (d) $w_{t+1} \sim \mathrm{Sample}(\mathbf{p})$ (choose next token)
   (e) $t \leftarrow t + 1$
3. **Until** end-of-sequence token or max length.
4. **Return** $w_{N+1:t}$.

The sampling strategy in step 2(d) controls the quality/diversity tradeoff:

Greedy decoding: Always take $w_{t+1} = \arg\max_k p_k$. Deterministic, but repetitive.

Top-k sampling: Zero out all but the k most probable tokens, renormalize, and sample. When $k = 1$, this reduces to greedy decoding.

Top-p (nucleus) sampling: Retain the smallest set $\mathcal{S}$ such that $\sum_{w \in \mathcal{S}} p(w) \geq p$, renormalize, and sample. This adapts the number of candidates to the shape of the distribution.

Parameter Count and Scaling

Proposition 9 (Parameter count per block)**.** *The number of non-embedding parameters per transformer block is* $12\,d^2$ *(assuming* $d_{ff} = 4d$ *and* $d_k = d_v = d/A$*).*

Proof. The attention sub-layer has four weight matrices:

$$\underbrace{\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V}_{\text{each } d\times d_k \text{ per head}} \text{ and } \mathbf{W}^O$$

With A heads and $d_k = d/A$, each of $\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V$ contributes $A \cdot d \cdot (d/A) = d^2$ parameters, and $\mathbf{W}^O \in \mathbb{R}^{d\times d}$ contributes d^2. Total for attention: $4d^2$.

The FFN has $\mathbf{W}_1 \in \mathbb{R}^{d\times 4d}$ and $\mathbf{W}_2 \in \mathbb{R}^{4d\times d}$, contributing $4d^2 + 4d^2 = 8d^2$.

Total per block: $4d^2 + 8d^2 = 12d^2$. □

For L blocks, the total non-embedding parameter count is:

$$N_{\text{params}} \approx 12\,L\,d^2$$

Example. GPT-3: $L = 96$, $d = 12{,}288$. Then $12 \times 96 \times 12{,}288^2 \approx 175$ billion parameters.

Kaplan et al. (2020) showed that the loss follows a **power law** in model size N, dataset size D, and compute C:

$$\mathcal{L}(N) \propto N^{-\alpha_N} \tag{12}$$

$$\mathcal{L}(D) \propto D^{-\alpha_D} \tag{13}$$

$$\mathcal{L}(C) \propto C^{-\alpha_C} \tag{14}$$

These **scaling laws** allow us to predict the performance of larger models from small-scale experiments.

Summary: Complete Forward Pass

Collecting everything, the forward pass of a decoder-only transformer language model is:

1. *Embed + Position* (§2–3):

$$\mathbf{x}_t = \mathbf{E}[w_t] + \mathbf{P}[t], \qquad \mathbf{X}^{(0)} = [\mathbf{x}_1; \ldots; \mathbf{x}_N] \in \mathbb{R}^{N\times d}$$

2. *Transformer Blocks* (§4–9), for $\ell = 1, \ldots, L$:

$$\mathbf{O}^{(\ell)} = \mathbf{X}^{(\ell-1)} + \mathrm{MHA}\Big(\mathrm{LN}\Big(\mathbf{X}^{(\ell-1)}\Big)\Big) \tag{15}$$

$$\mathbf{X}^{(\ell)} = \mathbf{O}^{(\ell)} + \mathrm{FFN}\Big(\mathrm{LN}\Big(\mathbf{O}^{(\ell)}\Big)\Big) \tag{16}$$

3. *Final LayerNorm*: $\mathbf{H} = \mathrm{LayerNorm}(\mathbf{X}^{(L)})$
4. *Output Head* (§10):

$$\mathbf{u} = \mathbf{H}[N]\,\mathbf{E}^\top \in \mathbb{R}^{1\times V}, \qquad \mathbf{p} = \mathrm{softmax}(\mathbf{u})$$